

My research interest lies at the intersection of Natural Language Processing (NLP) and large language models (LLMs), specifically focusing on developing methods to address fundamental limitations in how LLMs acquire, store, represent, and utilize knowledge. I am particularly invested in tackling biases and performance gaps in low-resource languages, a passion reinforced by my background as a Sri Lankan, where the two native languages (Tamil and Sinhala) remain severely underrepresented in LLMs. However, I view low-resource languages as an entry point to broader questions about LLM capabilities, and I am eager to build upon these insights to address similar fundamental challenges across different domains where LLMs show limitations in knowledge acquisition and representation.

This passion, combined with my academic excellence and research experience, has prepared and motivated me to pursue a PhD directly. I graduated with a perfect **4.0/4.0 GPA**, ranking as the **top student** in my cohort and outperforming all peers across three undergraduate programs in the Faculty of Computing and Technology's Class of 2024.

Research experience in LLMs My research journey began toward the end of my third year as an undergraduate, driven by curiosity to investigate the state of Sinhala in LLMs. As a native Sinhala speaker, I noticed persistent failures in LLMs for tasks like accurate translation and context-aware generation for Sinhala, which a subsequent literature review confirmed as stemming from severe under-representation in pre-training data.

As a student at a Sri Lankan government university (ranked third nationally), during the initial stages I faced substantial challenges, mainly from limited computational resources and scarce ongoing LLM projects. Undeterred, I initiated my own project under Prof. S.R. Liyanage's guidance. The project explored efficient adaptation methods for LLMs in low-resource languages, focusing on Sinhala offensive language detection. The challenging nature of this downstream task, specifically due to the linguistic complexities and cultural specificities of the Sinhala language as well as the scarcity of high-quality annotated datasets, drove this research direction.

I developed a two-stage fine-tuning approach, incorporating an intermediate pre-fine-tuning step with masked rationale prediction to integrate human feedback (rationale). I also fine-tuned several smaller variants of LLaMA 3 and Mistral models with task-specific fine-tuning. On the SOLD (Sinhala Offensive Language Detection) benchmark, both approaches surpassed preexisting baselines, including GPT-4o, with the two-stage finetuning approach achieving the best results. The findings demonstrated that reinforcing token-level understanding via human rationale enhances downstream performance in low-resource settings. After 14 months of dedicated effort, facing multiple ACL Rolling Review (ARR) cycles and addressing reviewer feedback, the work culminated in a paper [1] accepted at the **NAACL 2025 (SRW)**.

To expand my initial scope, I proactively sought international mentorship. I presented my initial work to Dr. Tharindu Cyril Weerasooriya (Research Scientist), securing his mentorship and supervision throughout the project. Additionally, I received further guidance from Prof. Marcos Zampieri (George Mason University) and Prof. Christopher Homan (RIT). While incorporating their guidance, I **independently managed all project aspects**, from experimentation and LLM pipeline implementation to manuscript writing. To overcome GPU access barriers, I secured **\$1000** in Google Cloud research credits with Dr. Weerasooriya's support, enabling me to expand my methodology from smaller models like XLM-R to larger ones such as LLaMA 3 (8B) and Mistral 7B.

Other Research Experiences Following my graduation in late 2024, I joined my department (Department of Software Engineering) as a Teaching Assistant, driven by my passion for teaching and desire to gain academic experience, and after a semester I got promoted to Lecturer (part of temporary academic staff). In this role, I worked collaboratively with Dr. S.P. Kasturi Arachi, the department head, to supervise several final-year undergraduates on their research projects. One of these mentorship experiences naturally evolved into a research collaboration on a case study applying machine learning to customer segmentation and churn prediction in Sri Lanka's maritime shipping industry. Beyond providing guidance

to the student, my involvement developed into a second-author role, where I contributed to implementing ML pipelines, designing and executing experiments, and co-authoring the manuscript. The work is currently under review at ICARC 2026 (International Conference on Advanced Research in Computing).

Future Research Agenda My NAACL work led me to identify a research problem I'm passionate about solving. The under-representation of many languages like Tamil and Sinhala prevents the true potential of LLMs from being realized, limiting their usefulness as tools for these underrepresented communities. This lack of equity affects millions of people, and through my observations, I recognized two critical challenges: (1) the severe under-representation of these languages in LLM pretraining data and (2) the scarcity of expert-level annotators for domain-specific tasks in low-resource languages. My long-term research goal is therefore to develop generalizable methods that directly address both challenges that will enable LLMs to achieve robust, equitable performance across multiple underrepresented/low-resource languages rather than just a handful.

As groundwork for my long-term research goal, I am eager to first explore how LLMs fundamentally acquire, store, represent, and utilize knowledge. By deepening my understanding of these mechanisms, I aim to leverage these insights to enhance model reliability and performance across diverse downstream tasks. For me, graduate studies represent the ideal setting for this foundational work, specially providing the collaborative environment necessary to deeply investigate these fundamental questions. I believe that building this foundational knowledge will enable my research to extend beyond low-resource languages to address broader challenges in LLMs.

To address the first challenge of severe under-representation in pretraining data, I want to explore methods for building LLMs that learn efficiently from limited data. Furthermore I am eager to explore methods of developing efficient and high-quality synthetic data generation techniques for low-resource languages using existing LLMs, ensuring the generated data captures cultural and linguistic nuances. Inspired by the work I carried out in my NAACL project, I also want to explore methods to effectively integrate human feedback across pre-training, mid-training, and post-training stages to amplify the impact of sparse expert annotations in low-resource languages. With this approach I aim to address annotator scarcity while enhancing LLM downstream performance and mitigating cultural biases.

Why University of Florida? I specifically chose the University of Florida because of its world-class faculty in NLP and AI and my interest in joining the Natural Language Processing & Culture Lab. I would be honored to work with **Professor Bonnie J. Dorr**, due to her expertise in multilingual NLP. I am also eager to work with **Professor Yuanyuan Lei**, as her research interests in knowledge-enhanced LLMs complement my goal of addressing bottlenecks in LLMs for low-resource languages, tackling data scarcity by integrating external knowledge. Furthermore, I'm interested in working with **Professor Sonja Schmer-Galunder**, whose research focus on AI Ethics and NLP would enable me to explore fairness and bias mitigation, which is essential for my goal of building equitable LLMs for low-resource languages.

After my PhD, I aspire to build a career in academia as a professor. My decision to join teaching immediately after graduating top of my class reflects my passion for education alongside research. I look forward to serving as a TA during my graduate studies and eventually mentoring students while conducting research that advances equitable AI for underrepresented languages and communities.

References

- [1] S. Haturusinghe, T. C. Weerasooriya, M. Zampieri, C. M. Homan, and S. Liyanage. "Subasa - Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala." *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 4: Student Research Workshop)*. Ed. by A. Ebrahimi, S. Haider, E. Liu, S. Haider, M. Leonor Pacheco, and S. Wein. Albuquerque, USA: Association for Computational Linguistics, Apr. 2025, pp. 260–270. ISBN: 979-8-89176-192-6.